

Methodological and Ideological Options

Untangling the underlying drivers of the use of single-use food packaging

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ABSTRACT

Global production and use of food packaging have steadily increased since the 1960s. Packaging waste is now considered one of the world's most urgent environmental problems. Most responses to the problem have focused on recycling or the introduction of biomass-based materials. Whilst useful, these approaches focus on managing the symptoms and do not address why the food system has become reliant on single-use packaging. In this paper we use systems method to understand the drivers for the food packaging problem. We apply the first two steps of the systems method to develop a Causal Loop Diagram representing variables and feedback loops that influence the use of food packaging. The modelling process has been supported by a narrative literature review and by the analysis of the behaviour over time. The results show that there are three main interlinked subsystems that drive food packaging use: the globalisation, the expansion of supermarkets and the constraints on households' time. In order to reduce food packaging, structural changes are required. Shifting away from a growth-driven food economy towards a degrowth economic model may reduce the need for food packaging. Redesigning and decentralising food systems are just some of the potential solutions.

1. Introduction

Packaging plays an important role in modern food systems. Food systems are defined as systems that 'encompass the entire range of actors and their interlinked value-adding activities involved in the production, aggregation, processing, distribution, consumption and disposal of food products that originate from agriculture, forestry or fisheries, and parts of the broader economic, societal and natural environments in which they are embedded' (Nguyen, 2018, p. 1). Food packaging provides several functions: it facilitates storage, handling, transport, distribution, protection and preservation of food, as well as providing a way of communicating product information (Gustavo Jr et al., 2018). A wide range of materials are used for food packaging, such as paper, metals (e. g. steel and aluminium), glass, wood and plastics (Emblem, 2012). Plastics represent 20% of all packaging materials by weight. Glass also represents a 20% all packaging materials by weight, however plastics are used to package 53% of all goods, while glass is used to package only 10% of all goods (Pongrácz, 2007). This makes plastics the dominant packaging material.

The global food market is estimated to be worth approximately US\$8 trillion per year and, in 2014, the global packaged food market reached

US\$2.4 trillion (Clapp and Scrinis, 2017). Despite its many benefits, food packaging has a range of negative impacts. The food sector is the largest user of single-use packaging, consuming up to 35% of global packaging production (Rundh, 2005). Almost 95% of food packaging goes to waste after a single use (WEF, 2016). The production, consumption and management of single-use food packaging poses a number of environmental threats. All packaging materials have varying negative impacts on the environment (Pongrácz, 2007), the focus of this study, however, is on food packaging use and not on food packaging materials. There is a mounting body of research highlighting the effects of plastics in marine, freshwater and terrestrial environments (Wagner et al., 2014; Li et al., 2016; Geyer et al., 2017). To resolve this problem, the global focus has often been on unmanaged plastic waste. However, even when adequately managed, waste still has a significant impact on the environment. For example, by 2050, emissions from solid waste are expected to increase to 2.6 billion tonnes of CO_{2e} per year (Kaza et al., 2018) and plastic packaging primarily uses fossil fuel feedstocks that have a significant carbon footprint. Globally, 6% of oil consumption is used for plastics (not just packaging), and by 2050, projections show that the plastic industry will account for 20% of total oil consumption and be responsible for 15% of the global annual carbon budget (WEF, 2016).

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Despite general awareness of the environmental impacts caused by food packaging, only incremental reductions in food packaging waste have been achieved. The literature on solutions to food packaging waste focuses mostly on recycling and waste management (Al-Salem et al., 2009; Eriksen et al., 2019) or on the engineering of alternative biomaterials (Siracusa et al., 2008; Madival et al., 2009; Han et al., 2018; Ainara et al., 2019). Whilst useful, these solutions do not rely on a systemic understanding of why the use of food packaging and waste is growing and how it can be most effectively addressed. To develop any effective solution, we first need to understand why our current food systems rely so heavily on single-use food packaging. To explore this question, we use a systems thinking approach to understand the dependence of modern food systems on packaging. Our objective is to initiate discussion on the need to rethink societal functions that drive the use of food packaging, such as family dynamics and the globalisation of the food market, and to consider what the best options are to reduce our dependence on food packaging.

The aim of this paper is to present a system wide perspective of the drivers that lead to the use of food packaging. First, we review the current solutions that aim at tackling the food packaging problem, with a special focus on recycling schemes and on biomass-based sustainable materials research. Second, we describe the systems method used to develop a dynamic hypothesis for the growth in food packaging use and waste. Third, we present the results of the application of systems methods, starting with problem articulation and followed by an explanation of our dynamic hypothesis split into three interrelated subsystems: globalisation dynamics, supermarket dynamics and household dynamics. Finally, we discuss our results and their implications for potential systematic solutions to the food packaging problem and the need for further quantitative research.

1.1. Current solutions to the food packaging problem

1.1.1. Recycling: curing the symptoms

Despite mainstream large-scale recycling being introduced globally 40 years ago, only 14% of plastic packaging is currently collected for recycling globally (WEF, 2016). Although most plastics are potentially recyclable, technological and economic constraints prevents most of them from being recycled. More than 80% of the plastics end up in landfill, incinerated or in the wider environment (Ignatyev et al., 2014; Brooks et al., 2018). Higher-income countries prefer to export their waste to lower-income countries, with polymer materials used for single-use plastic food packaging accounting for 89% of waste exports (Brooks et al., 2018). In 2016, countries of the Organisation for Economic Co-operation and Development exported 70% of their plastic waste to developing countries (Brooks et al., 2018). Even when recycling happens, the material value recovered and retained for subsequent use is only 5% globally (WEF, 2016). As Geyer et al. (2017) state recycling delays, rather than avoids, final disposal. This highlights the fact that recycling should not be the primary option used to tackle the food packaging problem.

1.1.2. New materials: a sustainable option?

Previous studies have compared the environmental footprint of alternative packaging materials, such as glass jars and plastic containers for baby food (Humbert et al., 2009), milk packaging made of a laminated foil of paper, polyethylene and aluminium (PA-PE-Al laminate) and polyethylene (Xie et al., 2011). The comparisons are based on a life cycle assessment (LCA), that generally only compares a few pre-existing packaging options against a small number of environmental performance indicators. For example, the results of Xie et al. (2011) concluded that if resource saving is the primary objective, PA-PE-Al laminate packaging is a better option, while polyethylene plastic packaging would be the better option for ecosystem protection. These simple comparisons of two or more pre-existing packaging options ignore systemic solutions to the packaging waste problem and proffer a false

narrative that sustainable choices exist between current packaging alternatives.

Within the circular economy literature, there is emphasis on finding new renewable packaging materials to substitute non-renewable materials, such as petroleum based plastics (Madival et al., 2009). For example, bio-polymers can be used to produce biodegradable polymer packaging (Siracusa et al., 2008). In 2005, the global market value of biodegradable polymers exceeded US\$145 million and has grown at a rate of 12.6% per annum to reach an expected market value of US\$260 million in 2019 (Auras et al., 2004; Mahalik and Nambiar, 2010). Some studies have applied LCA to compare packaging derived from fossil fuel with packaging derived from biomass, such as cellulose, starch (that requires enzymatic action to biodegrade), poly-beta-hydroxyalkanoates (PHB) and polylactide Acid (PLA) (Mahalik and Nambiar, 2010; Yano et al., 2014). Yano et al. (2014) found that biomass-based packaging could reduce GHG emissions by 14% to 20% compared to fossil fuel-based packaging, and if 100% recycled, biomass-based packaging could reduce GHG emissions by up to 32%. Gironi and Piemonte (2011) compared bioplastics (starch-based plastics) with plastics and found that while bioplastics could reduce the use of fossil fuels, the intensive corn cropping required to produce bioplastics could be detrimental to ecosystems. Biomass-based packaging and bioplastics may have the capacity to reduce the use of fossil fuels but, if the demand for packaged food products continues to grow, so will the demand for energy, water and raw materials (Andrews, 2015). When it comes to recycling, traditional plastics may produce fewer GHGs compared to the composting processes for the bioplastics (Gironi and Piemonte, 2011). It follows that the use of renewable packaging materials is a context-dependent issue and not always a better alternative (Koskela et al., 2014).

The market still lacks fully biodegradable plastic packaging materials. Some plastics are designed to be biodegradable, but in order to decompose they need the right exposure to light, oxygen and particular temperature conditions (Kaza et al., 2018). Even in cases where plastics can be biodegraded by bacteria and fungi, the degradation time remains mostly unknown (Andrady, 2011). There are plastics that are sensitive to photodegradation, which means ultraviolet light from the sun can lead to the fragmentation of the plastic into small pieces (Andrady, 2011; Li et al., 2016). However, in the ocean the rate of photodegradation slows (Andrady, 2011) and complete degradation may take over 50 years (Müller et al., 2001). Therefore, in order to activate the plastic degradation process, the right natural conditions have to be present. Today, most of the plastics used are resistant to degradation, and when they do degrade, they fragment in small microplastic particles that can create further environmental damage (Li et al., 2016).

If the food system does not fundamentally change then the danger is that it will simply transition from single-use plastic packaging to single-use biomass-based packaging, which is still resource intensive to produce and has a short life before disposal. A change in packaging materials will not change consumer behaviour, which is currently anchored in a throw-away culture. As discussed, recycling and the use of new materials are weak solutions to the food packaging problem. They are solutions that simply address the waste symptoms of food packaging rather than seeking to tackle the systemic causes of the problem. Understanding the systemic drivers of the growth in food packaging could lead to the redesign of the current food system with the goal of reducing the dependence on packaged food.

2. Methods: understanding the system influencing the use of food packaging

To tackle food packaging issues, it is important to understand the structural transformation of food systems from a socio-economic perspective. Therefore, we adopted systems method to identify the multi-dimensional aspects related to the use of food packaging. The specific steps used in a systems method are: (1) problem articulation, (2) dynamic hypothesis formulation, (3) simulation modelling, (4) model

testing, and (5) policy design and evaluation (Stermann, 2000). In this paper, we focus on the first two steps to develop a dynamic hypothesis that explains the growth in food packaging. In step one, problem articulation, the problem is defined as a dynamic problem using behaviour over time graphs for key system variables. These behaviour over time graphs were derived from historical data. In step two, dynamic hypothesis formulation, a Causal Loop Diagram (CLD) is used to present the feedback loops that explain the behaviours described in the problem articulation step. This paper focuses on the first two qualitative steps of systems method because the main interest is to understand the system structure responsible for food packaging growth. This process of applying the first two steps in a systems method has been used in several studies in different fields (Mai and Smith, 2015; Oviedo-Ocaña et al., 2016; Agnew et al., 2018; Sahin et al., 2020). To date, and to the best of our knowledge, studies exploring the system structure responsible for food packaging growth do not exist.

In this study, the CLD was developed through a narrative literature review of scientific peer-reviewed publications and reports from recognised institutions in the field, such as the United Nations or Government Agencies. The advantage of a narrative literature review (versus a systemic review) is that it is not limited by pre-determined search structure, allowing for a more broad and inclusive literature review (Berninger et al., 2020). In addition, a narrative literature review is useful when describing the history and development of a problem (Green et al., 2006). The literature reviewed encompasses work from different fields, such as economics (e.g. globalisation), food sciences (e.g. food packaging materials and functions), and social sciences (e.g. studies of family time and meals management). As presented more in-depth in the problem articulation (Section 3.1), the timeframe considered in this study was from 1960s to today, which is the period in which globalisation and food packaging production increased exponentially (Monteiro et al., 2013; WEF, 2016; FAO, 2017). The literature search was conducted using Google Scholar, PubMed and the University of Queensland research browser. Only papers in English were included in the review.

CLDs represent the causal links among variables in a system and how these causal links create feedback loops that influence the system behaviour over time. Cause and effect relationships between pairs of variables within a CLD are represented using arrows (the cause is at the base and the effect is at the head of an arrow) and each arrow has a polarity to represent the nature of the relationship. A plus ('+') indicates that cause and effect move in the same direction, while a minus ('-') indicates that cause and effect move in opposite directions. A feedback loop is created when a chain of cause-and-effect relations start at one variable and loop back to the same variable. Feedback loops are either reinforcing (represented by an R) or balancing (represented by a B) (Sherwood, 2011). Reinforcing loops act to amplify change over time (producing growth and decline), while balancing loops act to counteract or stabilise change over time (producing goal seeking behaviour or oscillations). The CLD presented in this paper is not designed to replicate the global food system. Rather its purpose is to highlight the main feedback loops that affect the use of food packaging. Stella Architect software (isee systems) was used to build the CLD.

In systems thinking the four levels of analysis are 1) events and outcomes that represent the current situation, which emerge from patterns and trends; 2) patterns and trends that represent the behaviour of the system over time, which emerge from system structures; 3) system structures that represent interactions, feedback loops and rules within the system, which emerge from mental models, and 4) mental models that represent beliefs, assumptions or world views, which represent the most in-depth level of understanding of the system (Maani and Cavana, 2007). The CLD helps visualise the system structure and provides insights into the mental models driving the use of food packaging.

3. Results

3.1. Problem articulation

In the study of system behaviour, three fundamental modes of behaviour are generally recognised. These are exponential growth, goal seeking and oscillation. These behaviours are generated by specific feedback loops, for example exponential growth is produced by reinforcing feedback loops, while goal seeking and oscillation are produced by balancing feedback loops (Stermann, 2000). The fundamental modes of system behaviour combine to produce more complex modes of behaviour, such as S-shaped growth, which are produced by combinations of reinforcing and balancing loops that shift in dominance over time. This relationship between system behaviour and structure allows us to identify feedback loops present within a system that is responsible for producing behaviours of the system. The behaviours of the system represent the problems to be addressed, while the feedback loops that explain the behaviours represent our dynamic hypothesis for why the problems occur. This section presents the behaviours of the food packaging system.

In this study, we considered the behaviour of the food packaging system from the 1960s to the present day, which is the period when: international trade of food products increased exponentially (FAO, 2017); supermarkets started to expand (Merrilees and Miller, 1997; Maxwell and Slater, 2004); the number of food products offered in supermarkets increased (Beitzen-Heineke et al., 2017); and the use of food packaging, in particular plastic packaging, increased exponentially (WEF, 2016).

Since the 1960s, the world has experienced an exponential growth in the international trade of agricultural products (Monteiro et al., 2013; FAO, 2017). In the 1980s and 1990s, international economic policies accelerated and intensified the international trade of food (Fig. 1a) (Monteiro et al., 2013; FAO, 2017). Over the same period, the adoption of labour-saving technologies by intensive farming agribusinesses led to a shift in labour demand from rural to urban areas. This resulted in rapid urbanisation (Fig. 1b) and a steady decline in the rural population (FAO, 2017; UN, 2018). Up until the 1970s, urbanisation affected mainly high-income countries, but the phenomenon is now widespread in low-income countries as well (FAO, 2017). The increase in the urban population translated into an increase in the number of people dependent on food purchasing (FAO, 2017). As a result, the role of supermarkets became more important in the food system (Konefal et al., 2005) and the size (Merrilees and Miller, 1997) and number of products offered by supermarkets grew rapidly (Fig. 1c) (Farmer, 2012; Beitzen-Heineke et al., 2017).

Since the mid-1960 there has also been an increase in the number of kilocalories (kcal) consumed per capita per day (Fig. 1d) (FAO, 2002; Vasiliska and Rechkoska, 2012). This has resulted from the increased consumption of highly processed packaged food. Investigating changes in annual per capita sales of ultra-processed food in 79 countries, Monteiro et al. (2013) showed that the sale of frozen food and snacks continued to increase in high-, upper- and lower-middle-income countries between 1998 and 2012. An additional factor that has influenced the increased demand for processed ready-to-eat meals has been the increase in time-poor workers (Regmi and Gehlhar, 2001; Emblem, 2012) resulting in a reduction in the time available to prepare meals. This has occurred with the increase in women participation in the workforce (Bittman and Rice, 2002).

The rapid growth in international food trade, urbanisation and the need to transport food to urban centres, the rise in supermarkets and the growth in demand for processed convenience food, has led to the rapid growth in packaged food. The bulk of this packaging is in the form of plastic and has contributed to the exponential growth in the use of plastics globally (Fig. 1e) (WEF, 2016; PlasticsEurope, 2018). Today, approximately 50% to 60% of the market share of all consumer packaging is used for food and beverage packaging (Olsnats and Kaivo-Oja,

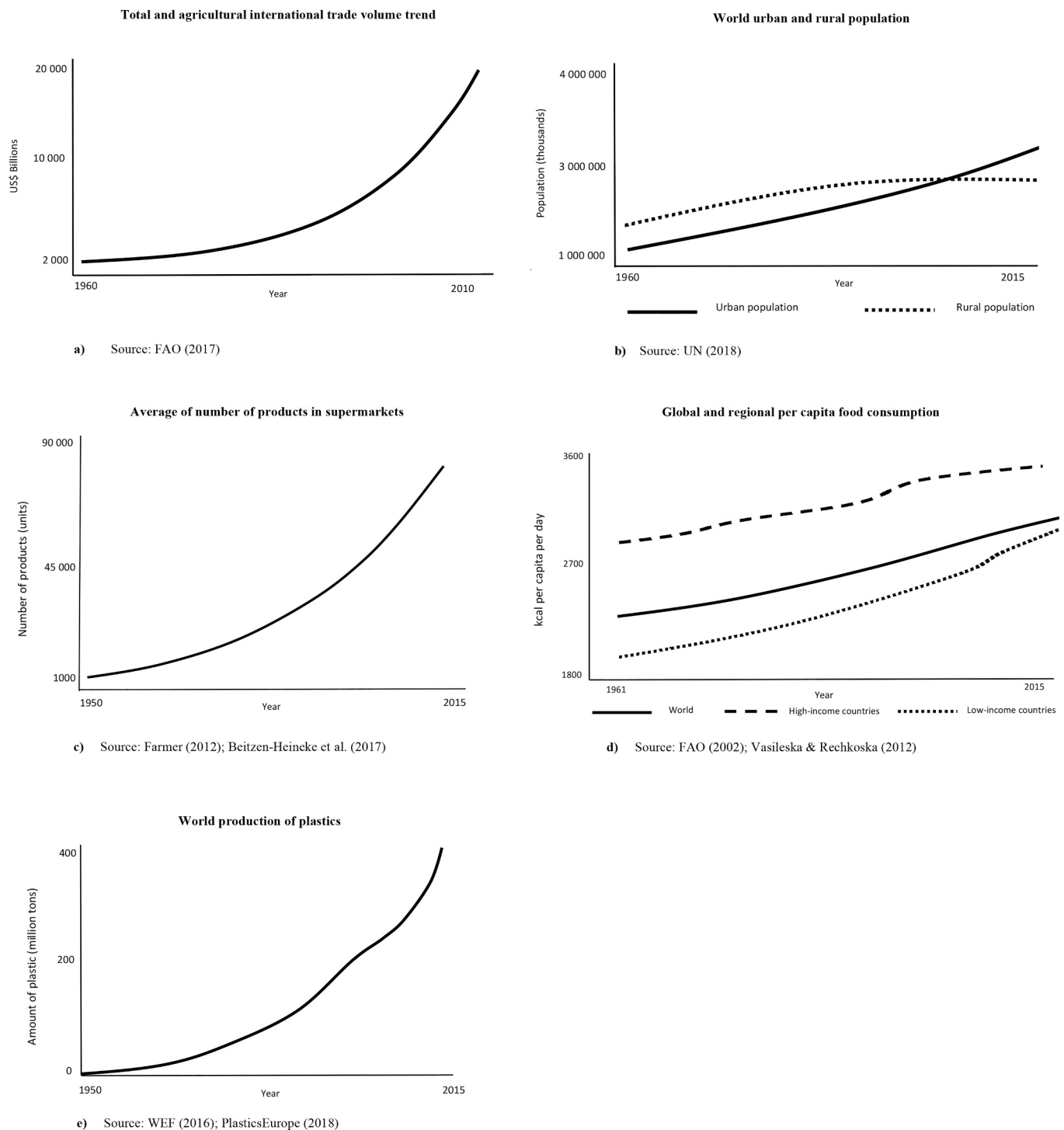


Fig. 1. Behaviour over time of key indicators in the global food system.

2014). The extensive use of plastics in packaging corresponds to an annual world production of approximately 200 MT, with an average per capita consumption of 100 kg (Mahalik and Nambiar, 2010).

3.2. Dynamic hypothesis: causal loop diagram

3.2.1. Food system dynamics

The complete dynamic hypothesis combining globalisation, supermarkets and household dynamics is presented in the CLD in Fig. 2. An explanation of the feedback loops within this dynamic hypothesis is presented in Table 1. The loops presented in Table 1 are intra-

subsystems (i.e. existing within one subsystem, e.g. loop R1 and R8) and inter-subsystems (including variables belonging to several different subsystems, e.g. loop R12 and R21), demonstrating the strong links between the systems. Only the three subsystems that influence the use of food packaging (Globalisation, Supermarkets and Households) have been presented in the results. The environmental and health impacts subsystem is not described in detail in this paper because it represents the symptoms of the use of food packaging, not the causes. A more complete CLD and a table presenting the references of the variables are included in the supplementary materials (Supplementary Fig. A1; Table A1).

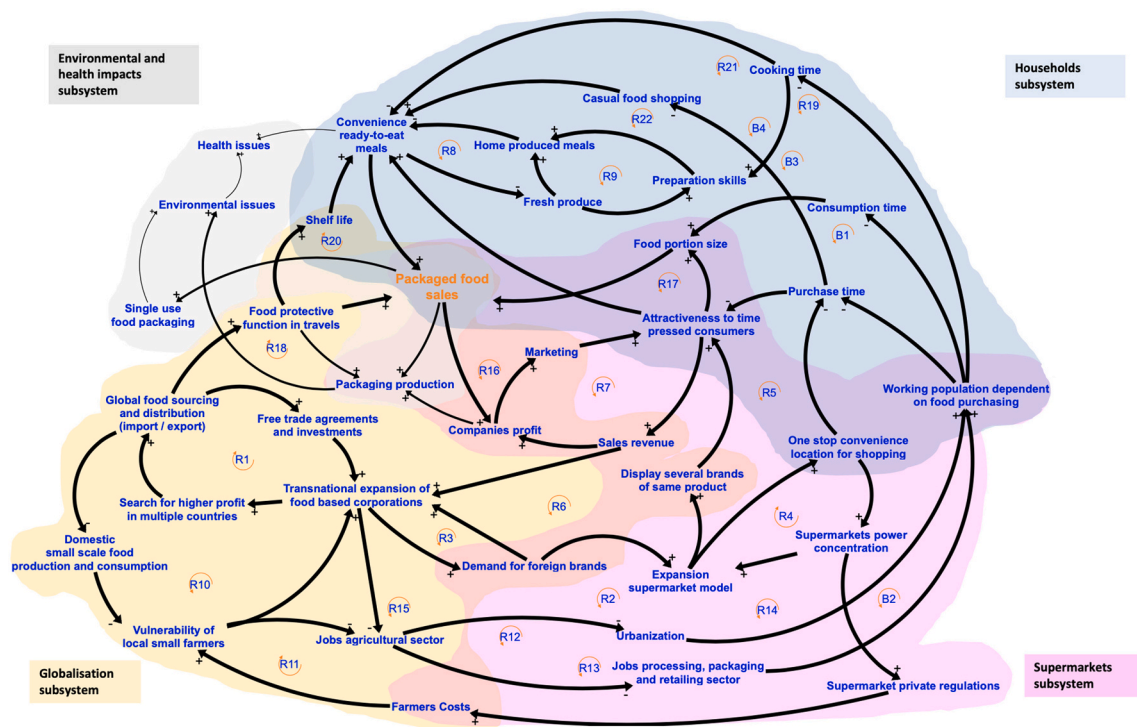


Fig. 2. Causal loop diagram of the drivers of food packaging use.

3.2.2. Globalisation dynamics

To understand the use of food packaging it is necessary to briefly look at the evolution of the global food system over the last two centuries. The most important transformation was the Industrial Revolution in the 19th century where food, which was traditionally sold in bulk, began to be sold as packaged products (Emblem, 2012). The new working class that moved to the cities meant that food previously produced and consumed at home now had to be transported to shops in urban areas (Emblem, 2012). The other major transformation was the globalisation of food systems in the 20th century. Globalisation is the process of greater interdependence and exchange of goods and services among countries. This mega-trend includes the exchange of knowledge, technology, production patterns and the homogenisation of consumer preferences and lifestyles (Qaim, 2017). Table 2 summarises the main differences between traditional food systems and modern global food systems.

The main transformation of food systems has been the shift in scale and geography (Maxwell and Slater, 2004; Konefal et al., 2005). Food globalisation has also been supported by the development of technology, lower transportation costs, tariff cuts and packaging innovations (i.e. fruit coating, bioengineering) that reduced the deterioration of perishable food (Regmi and Gehlhar, 2001). The global system of production and consumption, which is the dominant economic framework allowing food products to circulate internationally, has led to longer food supply chains (Sodano, 2008). The global food trade represents the first part of our dynamic hypothesis. The process of globalisation created reinforcing loops (loops R1, R2, R3 and R10, Fig. 3), in which growth in the import and export of food led to growth in the use of food packaging. These reinforcing loops correspond to the exponential growth trend presented in Fig. 1a (Section 3.1). Although elements of resistance to globalisation exist, demanding, for example food sovereignty (Francis, 2013; Tefft et al., 2017), globalisation dominates the food market. The increase of global trade increased the need for processing and packaging of raw material into food packaged products (loops R1 and R18, Fig. 3) (Erickson, 2008). As a consequence, with the increase in food trade, the protective function of food packaging became more crucial.

The main actors in the global food economy are transnational corporations (TNCs), which are also called 'Big Food' companies (Monteiro et al., 2013). The global market saw the development of food-based corporations, such as Nestlé, Unilever, Monsanto, Cargill and many others, seeking cheap labour and higher profits in multiple countries (Phillips, 2006). More generally, TNCs are transnational in their corporate constitution, multinational in sourcing their products, international in their labour exploitation, and global in their marketing approaches (Konefal et al., 2005). These companies benefited from the freedom to make foreign direct investments that have grown from \$US 61 million in 1985 to \$US 1.07 trillion in 2000, and that reached \$US 1.73 trillion in 2015 (Monteiro et al., 2018). By selling ultra-processed packaged food, these large corporations achieved an enormous commercial advantage over fresh and perishable products (Monteiro et al., 2013). Ultra-processed food is highly palatable, with a long shelf-life and convenient to consume anywhere at any time (Monteiro et al., 2018). Within the processed food products category, in 2002 packaged food accounted for \$1.1 trillion and beverages for \$641 billion (Gehlhar and Regmi, 2005). TNCs dominate the processed and packaging food sector (Clapp and Scrinis, 2017). Their profit is driven by the global trade of highly palatable food (due to high fat, sugar and salt content) and sophisticated and aggressive marketing strategies (such as reduced price for supersize servings) (Monteiro et al., 2013). Thus, their profit continues to drive the expansion of their market power, which translates into more global food trade in long and complex supply chains. TNCs branching into foreign countries increases food miles, out-competing local small producers with shorter supply chains (loops R10 and R11, Fig. 3) (Phillips, 2006; Sodano, 2008). As a result, at a macro level, packaged food evolved to become a key component in the globalised food system.

3.2.3. Supermarket dynamics

Urban areas are increasingly supplied with food via supermarkets (FAO, 2017). Urbanisation, which continues to increase (Fig. 1b), requires food to be transported and stored (FAO, 2017), and as a consequence, food products need to be packaged. Urbanisation has also led to

Table 1
Feedback loops within the causal loop diagram presented in Fig. 2.

Balancing loops	Variables
B1	Consumption time (+) Food portion size (+) Packaged food sales (+) Companies profit (+) Marketing (+) Attractiveness to time pressed (+) Sales revenue (+) Transnational expansion of food-based corporations (-) Jobs agricultural sector (-) Jobs processing, packaging and retailing sector (+) Working population dependent on food purchasing (-)
B2	Consumption time (+) Food portion size (+) Packaged food sales (+) Companies profit (+) Marketing (+) Attractiveness to time pressed (+) Sales revenue (+) Transnational expansion of food-based corporations (-) Jobs agricultural sector (-) Urbanisation (+) Working population dependent on food purchasing (-)
B3	Packaged food sales (+) Companies profit (+) Marketing (+) Attractiveness to time pressed (+) Sales revenue (+) Transnational expansion of food-based corporations (-) Jobs agricultural sector (-) Jobs processing, packaging and retailing sector (+) Working population dependent on food purchasing (-) Cooking time (-) Preparation skills (+) Home produced meals (-) Convenience ready to eat meals (+)
B4	Packaged food sales (+) Companies profit (+) Marketing (+) Attractiveness to time pressed (+) Sales revenue (+) Transnational expansion of food-based corporations (-) Jobs agricultural sector (-) Urbanisation (+) Working population dependent on food purchasing (-) Cooking time (-) Preparation skills (+) Home produced meals (-) Convenience ready to eat meals (+)
Reinforcing loops	Variables
R1	Transnational expansion of food-based corporations (+) Search for higher profit in multiple countries (+) Global food sourcing and distribution (import / export) (+) Free trade agreements and investments (+)
R2	Transnational expansion of food-based corporations (+) Demand for foreign brands (+) Expansion supermarkets model (+) one stop convenience location for shopping (+) Supermarkets power concentration (+) Supermarkets private regulations (+) Farmers costs (+) Vulnerability of local small farmers (+)
R3	Transnational expansion of food-based corporations (+) Demand for foreign brands (+)
R4	Expansion of supermarkets model (+) One stop convenience location for shopping (+) Supermarkets power concentration (+)
Reinforcing loops	Variables
R5	Expansion of supermarkets model (+) One stop convenience location for shopping (-) Purchase time (-) Attractiveness to time pressed consumers (+) Sale revenue (+) Transnational expansion of food-based corporations (+) Demand for foreign brands
R6	Expansion of supermarkets model (+) Display several brands of same product (+) Attractiveness to time pressed consumers (+) Sales revenue (+) Transnational expansion of food-based corporations (+) Demand for foreign brands (+)
R7	Marketing (+) Attractiveness to time pressed consumers (+) Sales revenue (+) Companies profit (+)
R8	Convenience ready to eat meals (-) Fresh produce (+) Home produced meals (-)
R9	Convenience ready to eat meals (-) Fresh produce (+) Preparation skills (+) Home produced meals (-)
R10	Vulnerability of local small farmers (+) Transnational expansion of food-based corporations (+) Search for higher profit in multiple countries (+) Global food sourcing and distribution (import / export) (-) Domestic small scale food production and consumption (-)
R11	Transnational expansion of food-based corporations (+) Demand for foreign brands (+) Expansion supermarkets model (+) one stop convenience location for shopping (+) Supermarkets power concentration (+) Supermarkets private regulations (+) Farmers costs (+) Vulnerability of local small farmers (+)
R12	Vulnerability of local small farmers (-) Jobs agricultural sector (-) Urbanisation (+) Working population dependent on food purchasing (-) purchase time (-) Attractiveness to time pressed consumers (+) Sales revenue (+) Transnational expansion of food-based corporations (+) Search for higher profit in multiple countries (+) Global food sourcing and distribution (import / export) (-) Domestic small scale food production and consumption (-)
R13	Vulnerability of local small farmers (-) Jobs agricultural sector (-) Jobs processing, packaging and retailing sector (+) Working population dependent on food purchasing (-) Purchase time (-) Attractiveness to time pressed consumers (+) Sales revenue (+) Transnational expansion of food-based corporations (+) Search for higher profit in multiple countries (+) Global food sourcing and distribution (import / export) (-) Domestic small scale food production and consumption (-)
R14	Jobs agricultural sector (-) Urbanisation (+) Working population dependent on food purchasing (-) Purchase time (-) Attractiveness to time pressed consumers (+) Sales revenue (+) Transnational expansion of food-based corporations (-)
R15	Jobs agricultural sector (-) Jobs processing, packaging and retailing sector (+) Working population dependent on food purchasing (-) Purchase time (-) Attractiveness to time pressed consumers (+) Sales revenue (+) Transnational expansion of food-based corporations (-)
Reinforcing loops	Variables
R16	Companies profit (+) marketing (+) Attractiveness to time pressed (+) Convenience ready to eat meals (+) Packaged food sales (+)
R17	Companies profit (+) marketing (+) Attractiveness to time pressed (+) Food portion size (+) Packaged food sales (+)
R18	Packaged food sales (+) Companies profit (+) Marketing (+) Attractiveness to time pressed (+) Sales revenue (+) Transnational expansion of food-based corporations (+) Search for higher profit in multiple countries (+) Global food sourcing and distribution (import / export) (+) Food protective function in travels (+)
R19	Packaged food sales (+) Companies profit (+) Marketing (+) Attractiveness to time pressed (+) Sales revenue (+) Transnational expansion of food-based corporations (-) Jobs agricultural sector (-) Jobs processing, packaging and retailing sector (+) Working population dependent on food purchasing (-) Cooking time (-) Convenience ready to eat meals (+)
R20	Packaged food sales (+) Companies profit (+) Marketing (+) Attractiveness to time pressed (+) Sales revenue (+) Transnational expansion of food-based corporations (+) Search for higher profit in multiple countries (+) Global food sourcing and distribution (import / export) (+) Food protective function in travels (+) Shelf life (+) Convenience ready to eat meals (+)
R21	Packaged food sales (+) Companies profit (+) Marketing (+) Attractiveness to time pressed (+) Sales revenue (+) Transnational expansion of food-based corporations (-) Jobs agricultural sector (-) Urbanisation (+) Working population dependent on food purchasing (-) Cooking time (-) Convenience ready to eat meals (+)
R22	Packaged food sales (+) Companies profit (+) Marketing (+) Attractiveness to time pressed (+) Sales revenue (+) Transnational expansion of food-based corporations (+) Demand for foreign brands (+) Expansion supermarket model (+) One stop convenience location for shopping (-) Purchase time (-) Casual food shopping (+) Convenience ready-to-eat meals (+)

a shift in the workforce towards off-farm jobs, such as food processing, packaging, transportation, wholesaling and marketing (loops R12, R13 and R15, Fig. 4) (FAO, 2017).

Modern supermarkets emerged in the mid-1980s, increasing in average floor area from 800m² to 4000m² in ten years (Merrilees and Miller, 1997). The larger supermarkets were developed to offer one-stop shopping destinations for time-poor workers (loops R4, R5 and R6,

Fig. 4) (Merrilees and Miller, 1997; Dixon, 2007). Over the past few decades, supermarkets have increased their retail power and they form the key players in the global food-retail sector (Burch and Lawrence, 2005). The larger number of urban consumers dependent on food purchasing increases the power of supermarkets (loops R2, R4, R5, R6 and R11 Fig. 4). Globally, the food retail sector is an oligopoly in which there is concentrated power (Konefal et al., 2005). In North America, for

Table 2

Differences between traditional and modern global food systems, adapted from Maxwell and Slater (2004).

Food system feature	Type of food system	
	“Traditional” food system	“Modern” global food system
Principal employment in food sector	In food production and primary marketing	In food processing, packaging, retail, and marketing
Supply chain	Short, local	Long, global
Food production system	Diverse, varied productivity	Homogenous crops, intensive, high inputs (agrochemicals)
Production farm scale	Family-based, small to medium	Industrial, large
Food purchased from	Small markets or local shops	Large supermarkets chains
Influential scale	Local to national	National to global
Type food consumed	Basic, fresh, seasonal	Processed, ultra-processed, absence of seasonality
Packaging	Low, Bulk food	High, Individualised packaged products with brand name
Family meal time	Longer preparation and consumption time	Shorter preparation and consumption time with increased use of convenience food

example, large supermarkets and convenience stores account for 93% of consumer purchases and in Europe they account for 55% (FAO, 2017). In Australia, the two major retailers are responsible for 70% of the food retailer market (Dixon, 2007). In high-income countries the presence of these corporations is well established, however, as a response to the increased demand for value-added commodities, processed and pre-prepared food, they are now expanding to low-income countries (Bye-lee et al., 2013). In developing countries, traditional grocery retailers are being replaced by modern supermarkets (Qaim, 2017). For example, between 1984 and 1993, 64,198 small shops went out of business in

Argentina, and between 1991 and 1995, 5240 small shops went out of business in Chile (Reardon and Berdegue, 2002; Maxwell and Slater, 2004). In South Africa, supermarkets control 55% of the food retail sector (Weatherspoon and Reardon, 2003; Maxwell and Slater, 2004). China has followed the same trend. The first supermarket opened in China in 1990, and by 2002 the country had over 53,000 of them (with 30–40% per year sales growth) (Hawkes, 2008). Large-scale domestic and foreign supermarket companies, such as Carrefours and Wal-Mart, expanded quickly in China (Hawkes, 2008).

The power of supermarkets directly influences the use of food packaging. Farmers, for example, have become subject to supermarket product standards and, as a consequence, local small and medium scale producers are slowly being excluded from the global food system, primarily because of the inability to absorb the cost associated with quality and traceability standards (loop R2, R10, R11, R12 and R13 Fig. 4) (Richards et al., 2013). Moreover, small and medium scale producers face the obstacles of delivering large quantities of products to respond to supermarket needs (Maxwell and Slater, 2004). This dynamic has resulted in the displacement of food production, manufacturing and consumption, leading to longer supply chains in which food packaging is essential.

In addition, as income increases, the demand for variety increases as well and this is reflected in the increase in the number of food products offered in high-income countries (loop R2, R3, R5, R6 and R11 Fig. 4) (Gehlhar and Regmi, 2005). With the establishment of supermarkets as a new type of food retailer in Europe and North America (Fearne et al., 2001), several brands of the same product are now displayed together and packaging has become an integral part of marketing strategies used to attract consumers (loop R7, R16, R17, B1, B2, B3 and B4, Fig. 4) (Farmer, 2012). Transnational corporations invest vast sums on advertising and marketing of their products (Monteiro and Cannon, 2012). The marketing displays used in modern supermarkets have led to an increase in the use of packaging as a means of selling and branding

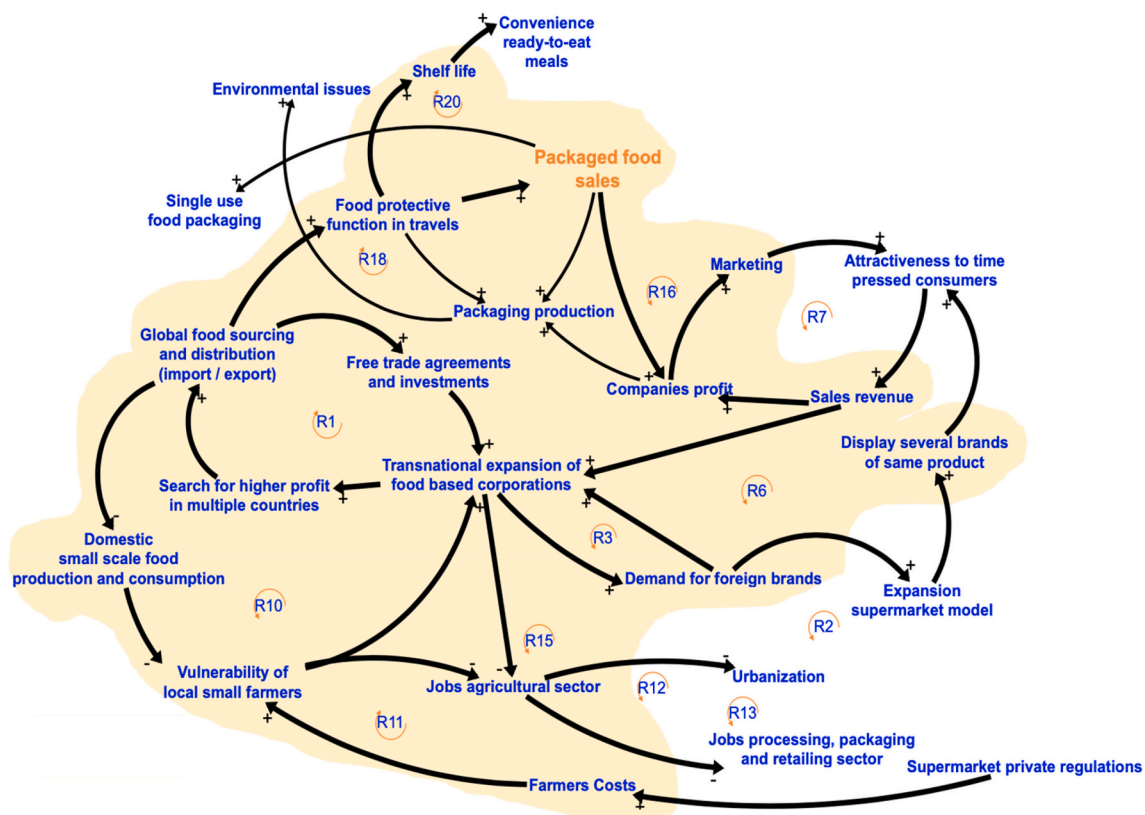


Fig. 3. Causal loop diagram: globalisation dynamics subsystem.

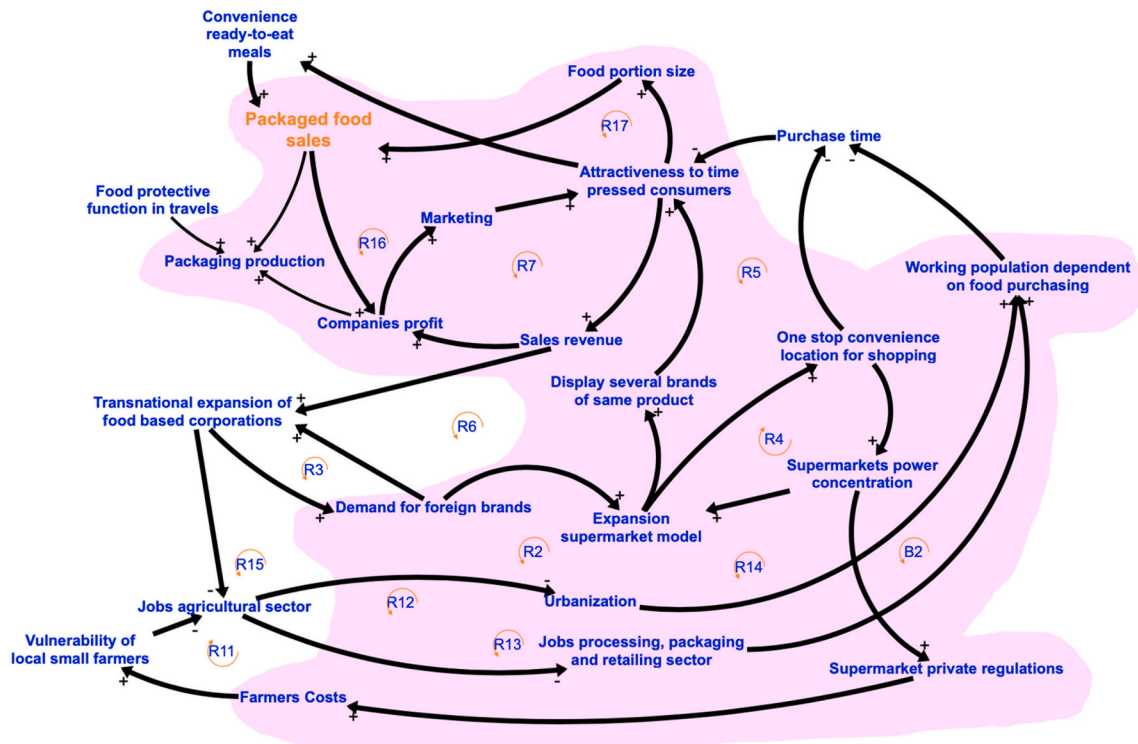


Fig. 4. Causal loop diagram: supermarket dynamics subsystem.

products to differentiate them from competing products (Marsh and Bugusu, 2007). In 1954, grocery shops stocked 1400 products on average and by 2015, this had grown to over 40,000 products in supermarkets across the world (Fig. 1c, Section 3.1) (Farmer, 2012; Beitzzen-Heineke et al., 2017). Packaging represents the first interaction that products have with consumers (loop R7, Fig. 4) (Marsh and Bugusu, 2007). Time-pressed consumers select products based on the subliminal cues of the packaging (Emblem, 2012). In the UK, for example, consumers spent an average of 29 s per product purchased on product selection (Grunert et al., 2010). Farmer (2012) reports that 7 s is the maximum produce attention time of consumers. Therefore, packaging has become a strategic marketing tool used by companies to increase their competitive advantage and improve their sales on overcrowded supermarkets shelves.

3.2.4. Households dynamics

Family meals have evolved over centuries, especially in terms of settings, duration, structure, social significance and participation in meal preparation and consumption (Ochs and Shohet, 2006). In parallel, agricultural trade shifted from bulk commodities (i.e. grains) to semi-processed (i.e. flour), processed (i.e. breakfast cereals) and ultra-processed products (Regmi and Gehlhar, 2001). Ultra-processed products are entirely or mostly made not from food, but from industrial ingredients and additives extracted or refined from whole foods (i.e. hydrogenated oils and fats, variants of sugar, flours and starches, cheap parts or remnants of animal products) (Monteiro et al., 2013). Examples of ultra-processed food products are frozen pasta and pizza, nuggets and sticks, chips, biscuits and cereal bars (Monteiro et al., 2013).

The demand for processed food is driven by the increase in purchasing power, increased opportunity cost of time required to purchase and prepare food and by the demand for higher quality and labour-saving products (Gehlhar and Regmi, 2005). In comparison to low-value food (fresh and bulk commodities), higher value foods (processed products) embody value-adding properties, such as reduced preparation time (Gehlhar and Regmi, 2005). Time-pressured people tend to adopt highly industrialised diets and rely more on packaged,

processed and pre-cut food (loops R8, R9, R16, R19, 20, R21 and R22, Fig. 5) (Phillips, 2006). This type of diet resulted in a worldwide increase in kcal consumed per capita per day (Fig. 1d, Section 3.1) (Vasileska and Rechkoska, 2012). Employment trends show that the workforce has become more casualised (casual and part-time job positions) and more insecure, with irregular work hours (Campbell, 1997; Bittman and Rice, 2002), while those with full-time employment tend to be overworked (Bittman and Rice, 2002). This has created irregular working schedules that reduce time for meal preparation and increase uncertainty around knowing who will be present for meals.

Research analysing family time in Belgium between 1966 and 1999 showed that the increase in dual-earner families reduced the time dedicated to family meals (Mestdag and Vandeweyer, 2005). Another study that focused on American parents showed that family meals are less regular when parents have non-standard work weeks (Golden, 2015). The problem is accentuated in single-parent families that face even greater time constraints due to competing work and family priorities. As a result, single-parent families have less time to prepare and participate in family meals (Mestdag and Vandeweyer, 2005; Fulkerson et al., 2006). Transnational companies and supermarkets that sell convenience, ready to eat food products leverage their benefits on the lack of time consumers have to prepare meals (loops R20, R21 and R22, Fig. 5).

Another aspect influencing family meals is the increase in employment of women with urban areas in particular having a greater percentage of women working outside of home (Regmi and Gehlhar, 2001). The increased opportunity cost of women's time has increased the demand for fast food (Regmi and Gehlhar, 2001) because women in many countries are still culturally and primarily responsible for preparing family meals. The growing reduction in time spent cooking also decreases confidence in food preparation (loops R8 and R9, Fig. 5), resulting in more children growing up without food preparation and cooking skills, making them more dependent on pre-prepared packaged food (Stead et al., 2004). An Australian study showed that involving children between the age of 10 and 12 in food preparation increased their confidence in cooking and in choosing a healthy diet containing

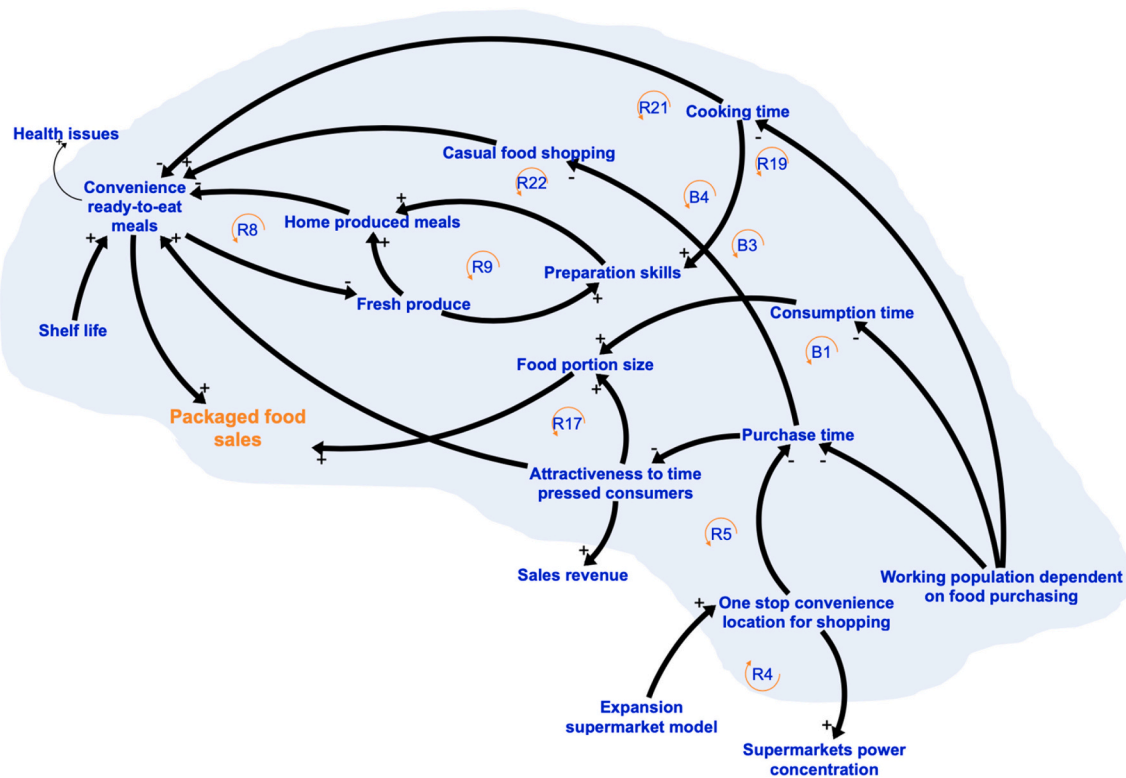


Fig. 5. Causal loop diagram: household dynamics subsystem.

less processed packaged food (loop R9, Fig. 5) (Leech et al., 2014).

Purchasing power and income elasticities are also important factors affecting family diets (FAO, 2017). Trends among countries of different income show that as income rises, consumers shift their diet towards more processed, packaged food products (Gehlhar and Regmi, 2005). A study on household environmental footprints in Australia revealed that higher income, time poor families seek the convenience of packaged food (Dey et al., 2007). On the other hand, low-income families are faced with barriers to purchasing healthy food (i.e. fruits and vegetables). International studies show that healthy food costs more compared to unhealthy food high in saturated fat, sugar and salt (Cade et al., 1999; Darmon et al., 2004; Jetter and Cassady, 2006; Kettings et al., 2009). Noble et al. (2007) found that lower income families in Australia and the UK had less access to fresh produce and a tendency to purchase large amounts of packaged frozen and processed food on a fortnightly or monthly basis.

4. Discussion: the need to rethink the system

4.1. Globalisation dynamics and food packaging

The problem of packaging remains a global challenge. Several countries have set a goal to phase out harmful packaging. For example, in March 2019 the EU announced a ban on 10 single-use plastic products that will come into force in 2021 (EU, EC, 2019). In Australia, the Federal Government works closely with the Australian Packaging Covenant Organisation to implement the 2017–2022 strategic plan to avoid and minimise the use of packaging (APCO, 2016). In several other Asian countries, recycling remains the most common response and some countries, such as Singapore, continue to invest in waste-to-energy technologies (UNEP, 2019). However, these types of public and private initiatives are just marginal solutions that tackle a post-consumption problem without considering the socio-economic drivers that influence the use of food packaging. The dynamic hypothesis presented in this paper has identified several reinforcing loops related to

globalisation (i.e. loops R1, R10 and R18, Fig. 3) that perpetuate the growth in food packaging use. Current policies that focus on the post-consumption management of food packaging waste will continue to be ineffective if these reinforcing loops are not addressed.

System structures emerge from mental models, which represent the beliefs, assumptions and world views of people within the system (Meadows, 2015). The current global food system remains fuelled by mental models that pursue capitalist economic growth, in which governments are focused on increasing Gross Domestic Product (GDP), and TNCs seek to increase profit (i.e. grow capital). The major neoliberal policies and institutional changes that influenced trade liberalisation of food products happened during the 1980s and 1990s (Monteiro et al., 2013; FAO, 2017). Harvey (2007) defines neoliberalism as the theory of political economic practice that proposes that human well-being is achieved by liberating entrepreneurial freedoms by establishing free markets and free trade. Free trade agreements encouraged global sourcing and distribution of goods across borders to increase profitable investments (Brand et al., 2020). Some recent large regional trade agreements that affect the trade of agricultural products include the Trans-Pacific Partnership (TPP), the Regional Comprehensive Economic Partnership (RCEP) and the Transatlantic Trade and Investment Partnership (TTIP) (FAO, 2017). These neoliberal open-market agreements fuel the capitalist growth model (Brand et al., 2020) by normalising TNCs' access to cheap resources and labour in the agri-food system. Hence, changes in policies should reconsider the size of the global food market. Rethinking the scale of food systems would be a first step towards reducing the use of food packaging and the resulting environmental impacts. To enact a system change, mental models (e.g. ideologies, values, myths and dogmas) need to shift. In fact, mental models are the greatest leverage points within a system (Meadows, 2015). Leverage points are the places in which small changes could lead to big improvements in the system (Meadows, 2015). Changing mental models would mean designing food systems that prioritise positive social and environmental outcomes instead of pursuing economic growth.

Reinforcing feedback loops drive growth and a system with

unchecked reinforcing loops destroys itself (Meadows et al., 2004; Meadows, 2015). Slowing growth, by reducing the dynamics of positive loops, is usually a more powerful leverage point than strengthening negative loops with marginal solutions (Meadows, 2015). New economic frameworks are emerging that aim to break the unchecked cycle of growth driven by reinforcing loops (D'Alisa et al., 2014). Economic degrowth is one of these, intended as the downscaling of economic metabolism, corresponding to a reduction of the flow of energy and materials in production and consumption (Schneider et al., 2010; Gomiero, 2018). In an economic degrowth society, social and ecological well-being, at the local and global level, would be prioritised over profit and GDP growth (Schneider et al., 2010; Kallis et al., 2012; Asara et al., 2015). Jumping off the treadmill of the growth-driven globalisation and re-localizing food systems, tackling therefore loops R1, R10 and R18, would be the first great leverage point that requires the attention of decision makers and key negotiators. The adoption of policies that promote economic degrowth could allow a shift away from the homogenisation of products, intensification of food production, and the growth in food processing and packaging.

TNCs are the main actors in the global food system. The annual turnover of some of these food-based corporations matches the GDP of middle-income countries (Monteiro and Cannon, 2012). Reinforcing loop R1 within the globalisation subsystem (Fig. 3) shows how with higher profits, these TNCs increase the trade of their products and, as a consequence, more food enters the global market, further increasing their market power. With a goal to further expand the food market to increase economic growth, products have also been exported and imported within the same industry, a mechanism called intra-industry trade, which has led to an expansion of trade that is not fuelled by growth in consumption (Regmi and Gehlhar, 2001). An example of intra-industry trade can be seen in the US, which exports higher valued beef to Japan while importing lower valued beef from New Zealand (Regmi and Gehlhar, 2001). Food systems that were once dominated by local, regional and national networks have now been globalised with an increased concentration of TNCs market power (Konefal et al., 2005).

TNCs vertically integrated all the processes within a global value chain, including primary production, large scale processing, manufacturing, distribution, and retail (FAO, 2017). To create and maintain the global agri-food market power these companies have developed their own private standards, as strategic instruments to regulate producers (Reardon and Farina, 2001; Bingen and Siyengo, 2002; Konefal et al., 2005). These private regulations, which are a dominant mechanism of control, as well as those enforced by food retailers (supermarkets), have resulted in a trend of increased exclusion of small scale farmers (i.e. loop R10, R11 and R12, Fig. 3) (Phillips, 2006; Swinnen, 2015). Due to the strong control that vertically integrated food corporations have, small scale farmers are faced with challenges such as market accessibility, financing, transport, as well as barriers related to quality, certification and traceability (FAO, 2017). The neoliberal capitalist transformation of the agri-food system has been driven by corporate's interests to increase profit (Konefal et al., 2005; Allen, 2010; Francis, 2013) and it is the root of the system (i.e. mental models) that needs to be changed in order to reduce the use of food packaging and alleviate its negative impacts. Solving this issue will be a challenge due to the strong market power of TNCs which allows them to shape regulatory frameworks in which they operate to further increase their power and market expansion (Clapp and Scrinis, 2017). If countries continue to support the globalisation of the food market, accommodating the monetary interests of TNCs, with the goal to achieve infinite economic growth, food packaging will continue to be produced and disposed of. Decentralising food systems and shortening the supply chains are appealing solutions because, in a globalised food system, food production, packaging and transportation are processes that consume large amounts of fossil fuel (Sodano, 2008), further contributing to a rapid depletion of the planet's natural resources. The longer supply chains also results in an increase in GHG emissions, congestion, accidents and noise

(Sodano, 2008).

4.2. Supermarket dynamics and food packaging

The supermarket subsystem represents the meso-level (national level) that intersects and connects the macro-level of globalisation (international level) to the micro-level of the household (families and individuals' level). Supermarkets have played an important role in the industrialisation and commercialisation of food systems (Maxwell and Slater, 2004). The expansion of supermarket power has led to a monopsony in distribution, which means that a large number of processing businesses are forced to sell their products to a limited number of globally-focused retailers that have strong purchasing power in a concentrated market (Burch and Lawrence, 2005). Supermarket supply chains require producers to provide large volumes of product (Timmer, 2014), which often requires intensive agricultural practices dependent on chemical fertilisers and fossil fuel energy. The use of these inputs has a significant global environmental impact (FAO, 2017). Moreover, under the influence of supermarket oligopolies, the food production system evolved from a large number of small producers to a small number of large producers (Konefal et al., 2005) and the large producers may not be close to the point of consumption. This necessitates longer supply chains and packaging.

Around the world there is a growing movement of zero-packaging grocery stores that encourage consumers to reuse and refill their containers, while providing a level of environmental education (Beitzen-Heineke et al., 2017). From a microeconomic degrowth perspective, Community Supported Agriculture (CSA) projects that prioritise local, and often organic products could be a potential substitute for supermarkets (Bloemmen et al., 2015). The CSA model offers a direct relationship between consumers and producers, which shortens the distances between consumers and producers and reduces the number of the intermediaries. Consumers are assured of a fair price for their local food, producers can balance their costs and revenues, rather than producing to maximise profits, and the ecosystems carrying capacity is respected (Bloemmen et al., 2015). These are simple examples of existing alternatives, but a detailed discussion is beyond the scope of this article. Re-localizing markets, re-territorialising production, re-seasonalising food consumption (Amate and De Molina, 2013) and decentralising food systems, may be the only way to shorten food chains, to reduce shelf competition (and related marketing strategies) and decrease the need of food packaging. These are all changes that could impact the dynamics of centralised food systems, represented by, for example, reinforcing loops R2, R4 and R7. Shifting the mental models towards an economic degrowth framework would allow to accelerate the implementation towards these goals.

4.3. Households dynamics and food packaging

Urbanisation underpins changes in food consumption patterns, as well as changes in the global food market. Today, approximately 54% of the world population lives in urban areas, and by 2050 this will increase to more than two-thirds, which could add 2.4 billion people to urban areas (UN, 2018). Changes in the total population affect the demand for food, however changes in population distribution will affect food consumption patterns. Global trends show that urban dwellers with high income prefer food products that reduce preparation time, increasing the demand for processed food and fast food (FAO, 2017). Packaged food is a proxy for processed and ultra-processed products (Monteiro et al., 2013). Beyond the environmental effects caused by longer supply chains and food packaging pollution, packaged processed food negatively affects human health too. The shift towards cheaper processed food has resulted in an increase in caloric availability, since convenience food is high in fat, added sugar and salt (also called 'empty calories') (FAO, 2017). The literature on the health effects of convenience food is extensive (Rauber et al., 2015; Monteiro et al., 2018), with clear

evidence that convenience processed food has increased the rate of obesity, diabetes, high blood pressure and heart disease in society (Monteiro et al., 2013). Born to counterbalance globalisation dynamics, in which standardised fast food are offered by centrally controlled TNCs, the Slow Food movement promotes alternative food practices that prioritise natural, local, traditional and sustainable food choices (Sassatelli and Davolio, 2010). By defending the use of primary local ingredients, Slow Food practices could help achieving a healthier diet and minimise the need of food packaging.

In addition, the household subsystem (Fig. 5) highlights how time (i.e. cooking time, shopping time, loops R19 and R21) is a crucial component affecting food consumption patterns and the choice of food products. In the dynamic hypothesis, all variables related to the lack of time influence 'convenience ready-to-eat meals', which increases the sale of packaged food. In households that have less time for cooking, there is an increased preference for highly processed convenience food (Regmi and Gehlhar, 2001). Knowledge about the ecological and health impacts of processed packaged food exist but, improving knowledge may be insufficient to induce a behavioural change. In contrast to the Slow Food movement, which claims that consumers are able to influence market relationships using their power (Sassatelli and Davolio, 2010), we believe that society's transformations are co-led by people and institutions, and it remains challenging to expect consumers to take the entire burden of living an ecologically sustainable lifestyle while also meeting their responsibilities and aspirations in the current growth-driven system. Within the degrowth paradigm, free time should be prioritised. People, for example, could work four days a week (Kallis, 2013) using work sharing mechanisms (Kallis et al., 2012), increasing the time availability to grow, cook and consume food. Therefore, the main leverage for change in this subsystem would be to introduce policies able to increase time availability, allowing people to shift away from the need to purchase packaged food (e.g. loop R17, R19 and R22). Shorter, more flexible working weeks could enable healthier, less-processed, packaging-free diets, breaking the existing patterns of several loops (e.g. R12, R13, R14, R15, and R21, B1, B2, B3, B4).

5. Conclusion

Recycling and the use of new biodegradable materials remain weak solutions to the food packaging problem. Government policies support the global food market and implement weak sustainability strategies, such as managing post-consumer waste. The dynamic hypothesis presented in this paper highlights three main subsystems driving the use of food packaging in the current global food system. At a deeper level, economic growth, as an ideology and economic imperative, has to be questioned. Has economic progress produced a perverse result in food systems? The corporate concentration in agribusinesses, retailing and distribution has led to longer and vertically integrated value chains. Thus, food packaging represents a symptom of the resource intensive global food economy. The food packaging problem is a food system problem, not a packaging problem. The food packaging problem is underpinned by a need to solve deeper societal issues, such as the pursuit of continued economic growth on a planet with limited resources, and the lack of time that modern families have to grow, purchase, cook and consume food, due to socio-economic pressure.

Processed packaged food harms the natural environment and human health. To reduce the use of food packaging it is necessary to develop food systems in which social and ecological well-being are prioritised over increased economic growth. Redefining the current economic model will be necessary. A transition to a degrowth economy might be required in order to achieve re-localisation of food production (i.e. investing in local farmers and supporting urban farming), a reduction in food transport and a transition away from convenient processed food. This study proposes a change in the economic paradigm that governs food system, it is however, beyond the scope of the study to suggest specific policies to implement an economic degrowth framework.

Understanding a problem from a systems perspective informs the development of policies aimed at changing the structural nature of the system in order to generate long-term solutions. Future research should focus on the development of system dynamic simulation models that allow these structural changes to be tested. These models would fill a gap within the degrowth literature, that still lacks simulation and testing of alternative food system transition scenarios. The goal would be to identify leverage points in the current global food system and to empirically test a range of policy options that could allow a shift towards a reduction in food packaging within a new, more socially just and environmentally sustainable economic model.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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